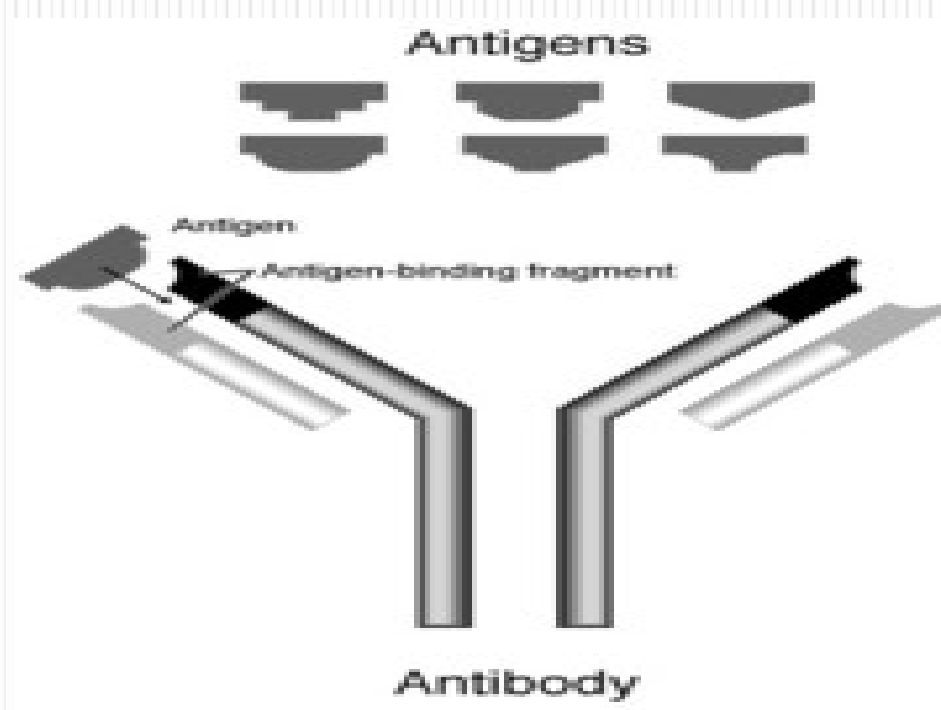


Antibodies



Learning objectives

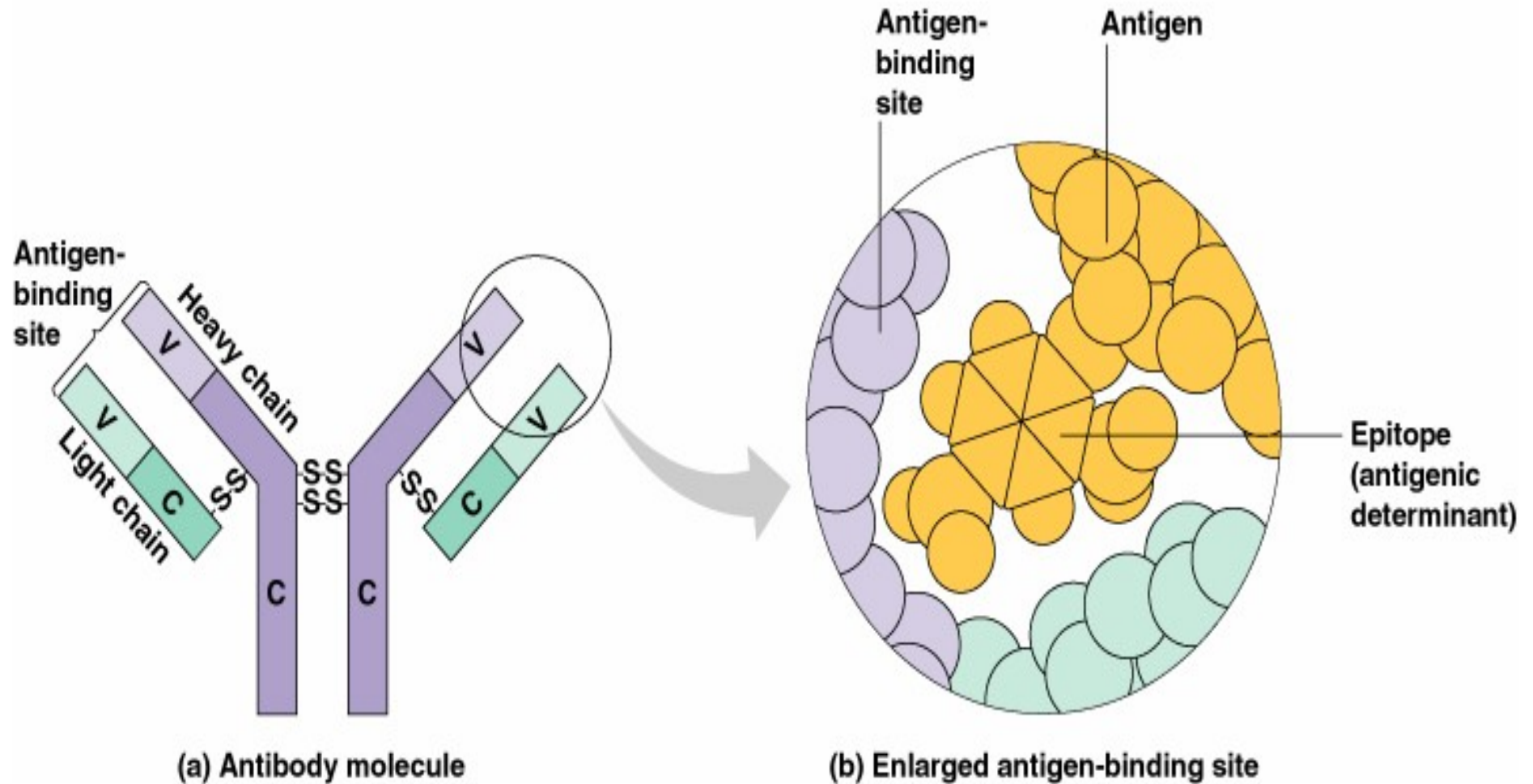
At the end of this session, student should be able to:

- ✓ Describe the different types of immunoglobulins and their main function
- ✓ Define monoclonal and polyclonal antibodies.
- ✓ Differentiate the types of mAbs
- ✓ Describe hybridoma technology.
- ✓ Describe the different application of mAbs.
- ✓ Understand the MoA of antibody as therapeutics.
- ✓ Mention components of mAb-drug conjugates and their mechanism of action.
- ✓ Describe the basic principles of ELISA test and its procedures.

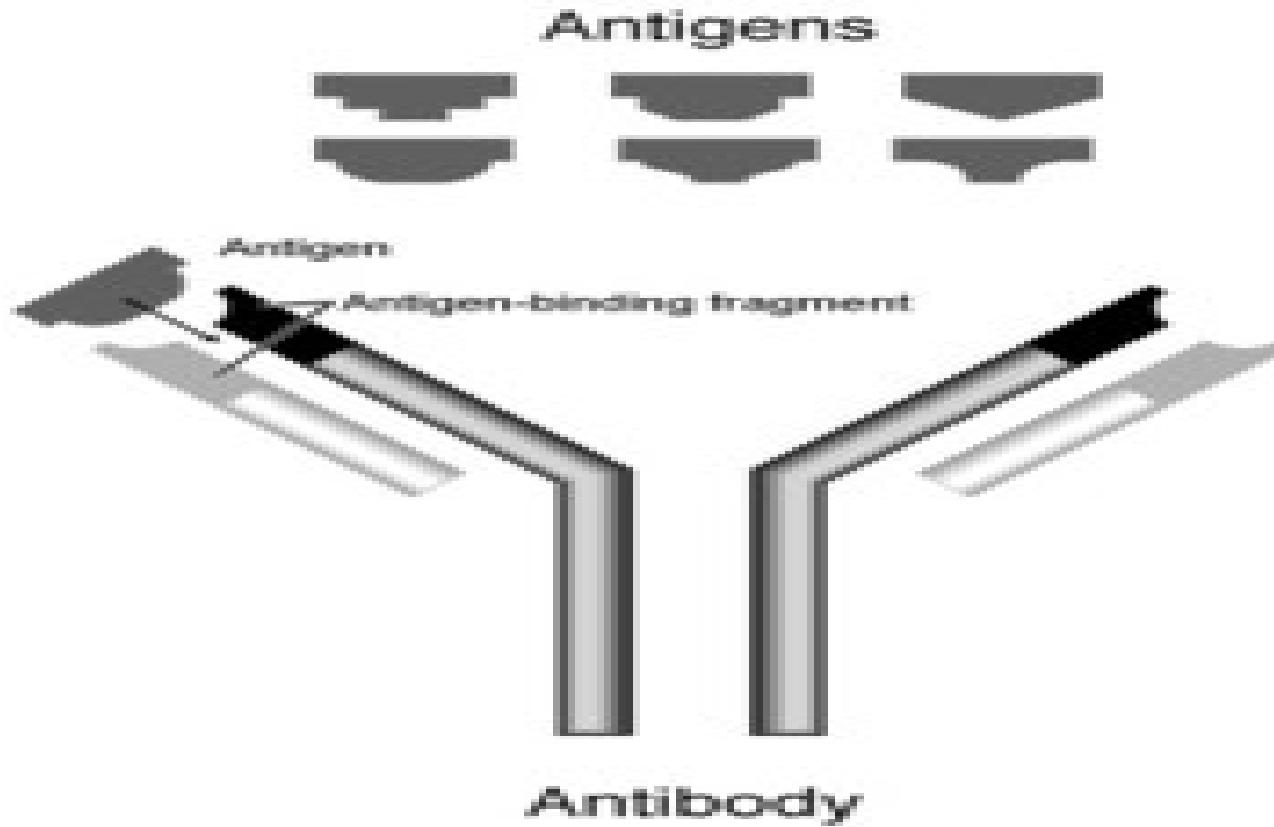
Antibodies

- ✓ Belong to a group of serum proteins called immunoglobulins (Igs).
- ✓ **Recognize and bind to a particular antigen with high specificity.**
- ✓ An antibody is a Y-shaped molecule with four protein chains:
 - 2 Identical light chains
 - 2 Identical heavy chains

Antibody structure



Antibody structure



Antibody structure

❑ Variable regions:

- ✓ Two sections at the upper part of Y's arms.
- ✓ Contain the antigen binding sites.
- ✓ Identical on the same Ab but vary from one Ab to another.

❑ Constant regions:

- ✓ Stem of monomer and lower parts of Y arms

Antibody structure

- Variability of antibody variable domains is largely restricted to three 'hypervariable' regions in each of the heavy and light chain variable domains.
- **These hypervariable regions, also known as complementarity determining regions, or CDRs, form loops at the tip of the Fab structure.**

Antibody structure

- The **hyper variability of these loops** is therefore responsible for the **different binding specificities of different antibody molecules**.
- The CDR loops vary in **both amino acid sequence and length** between antibodies.

Antibody structure

- The constant regions maintain a common structure of the molecule.
- Determines the mechanism used to destroy antigen.
- The constant regions allow interaction with the effector systems such as complement binding and binding to Fc receptors **on macrophages** to activate phagocytosis.

Function of antibody

Antigen–antibody binding leads to:

1. Agglutination

Abs cause antigens (or microbes) to clump together

2. Neutralization

Antibodies bind to pathogens (e.g., viruses, bacteria) and block their access to cells, then the antibody-antigen complex is engulfed by a macrophage.

Function of antibody.....

3.Opsonization:

- **Encapsulated bacteria** (e.g., Hib, pneumococcal, and meningococcal) evade the innate immune system because they are not recognized by macrophages or dendritic cells.
- However, encapsulated bacteria can be recognized by antibodies.
The antibody coats the bacteria to enable the ingestion by macrophages and dendritic cells through the process of opsonization

Function of antibody...

4. Complement activation

- Abs trigger the complement system
- Antibodies bind to certain bacteria in the plasma. **A region on the antibody is a receptor to complement proteins which will help lyse the bacteria or attract the macrophages to it.**

5. Precipitation- If the antigen is soluble, the reaction with specific antibody under appropriate conditions results in **precipitation of large antigen-antibody aggregates.**

Classes of Antibodies

Antibodies as a class are known as immunoglobulins(**Ig**)

✓ **IgG, IgM, IgA, IgD, IgE**

■ They differ in their physicochemical (charge, size, and solubility) and serologic (*in vitro reactions* with antigens).

IgG

- The most abundant Ig in serum.
- Constitutes about 80% of the antibody in serum.
- **The major antibody obtained after repeated immunization.**
- Produced slowly upon primary exposure to an antigen
- Produced rapidly during secondary or subsequent exposure, becoming the major antibody present
- **The principal humoral components of Immunological memory**
- The only antibody that crosses the placenta. It helps protect the newborn from infection through passive immunity.

IgM

- Serum IgM antibodies are pentamers composed of five monomeric IgM units (each consisting of two heavy and two light chains).
- Because they are multivalent, IgM antibodies are **better cross-linkers and agglutinators**

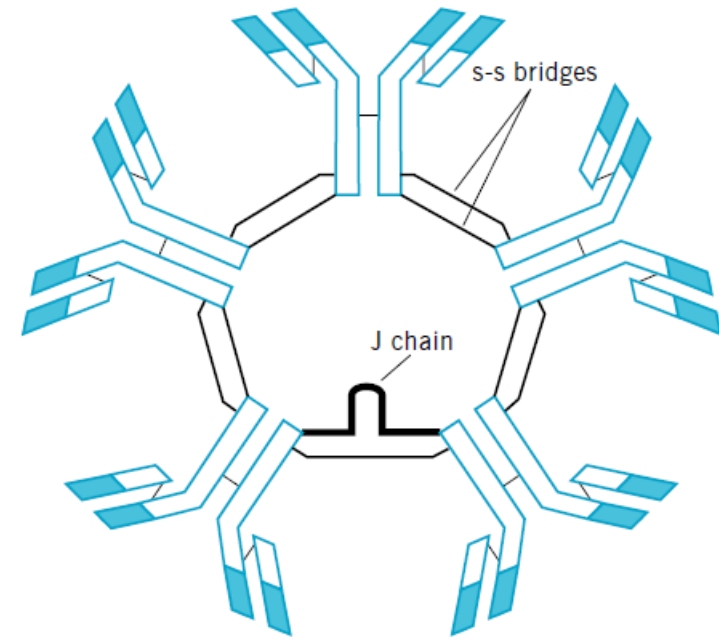


FIGURE 4-10 Human IgM. This is a pentameric (five

IgM.....

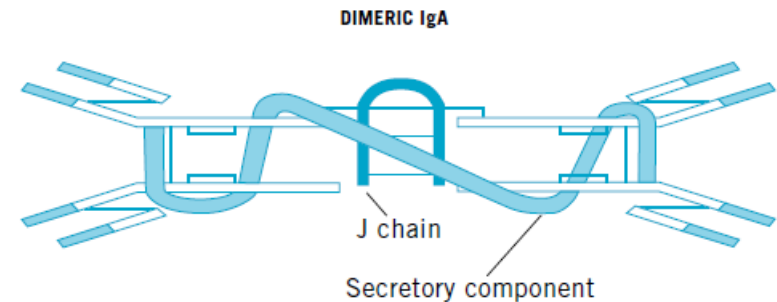
- A valuable diagnostic marker for infectious disease because it is usually the **first immunoglobulin made following Ag exposure and is relatively short-lived.**
- Effective in activating complement
- Participates in the lysis (bursting apart) of cells
- Generally remains in the blood; does not diffuse into the surrounding tissues due to its large size

IgE

- The rarest class of antibodies, appear in the serum in minute amounts (0.003%).
- Responsible for the allergic reactions mediated by binding to specific Fc receptors on mast cells.
- Elevated in people with hypersensitivity to allergens, as well as those with eczema, asthma, or other respiratory problems

IgA

- The main secretory immunoglobulin; found in exocrine secretions (e.g., breast milk, saliva, tears, respiratory and digestive secretions, urine).
- In the serum, IgA is found in both monomeric and polymeric form and in secretory fluids mainly in dimeric form.



IgA

- Represents approximately 10% to 20% of the immunoglobulins in serum
- Most abundant immunoglobulin in tissues
- **Prevents or interferes with the attachment of viruses and bacteria to mucosa of respiratory and digestive systems**
- Protects against enterotoxins released by certain bacteria; for example, forms an antibody-antigen complex with **cholera toxin**, preventing it from binding to specific receptors on the intestinal membrane.
- Plays a role in eliminating food antigens from the circulatory system.

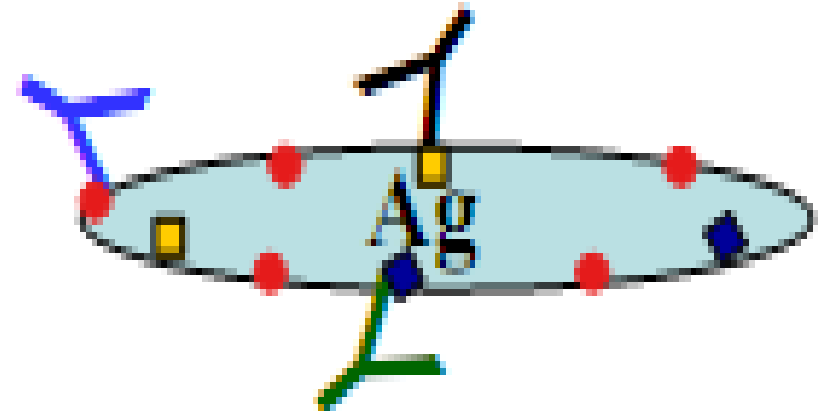
IgD

- IgD constitutes less than 1% of the antibody in human serum.
- **Acts as an antigen receptor on the surfaces of mature B cells**
- **Signal the B cells to be activated.**

Polyclonal and Monoclonal Antibodies

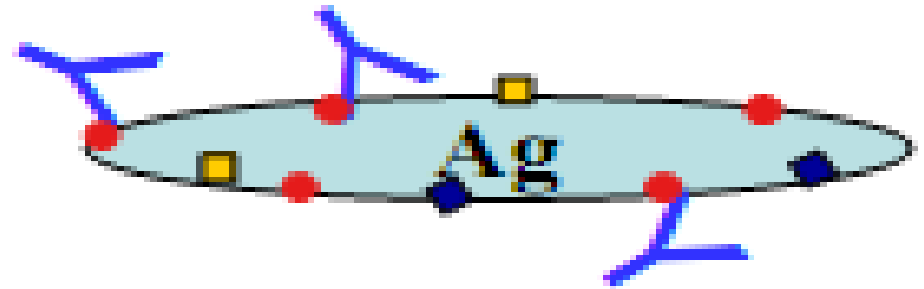
Polyclonal Antibodies

- ✓ mixture of antibodies with different specificities,
- ✓ product of different B cell clones,
- ✓ recognize different **epitopes** of an antigen



Monoclonal Antibodies (MAbs)

- Monoclonal antibodies are derived from a single ('mono-') clone of antibody-producing B cells.
- They are antibodies with the **same specificity**, able to recognize only **one epitope**.
- produced by a single source or clone of B cells.



Monoclonal Antibodies (MAbs).....

- Antibodies have important uses beyond fighting infections in the body.
- Production of long-lasting monoclonal antibodies is a recent invention and it is used in both medicine and research.
- **Monoclonal Antibody:** a stable antibody which can be used over a period of time

Production of monoclonal antibodies

Hybridoma technology

- Method in which large quantities of monoclonal antibodies are produced.
- Done by **administration of antigen** in mammal (usually mouse) which produces an immune response.

mAb production steps

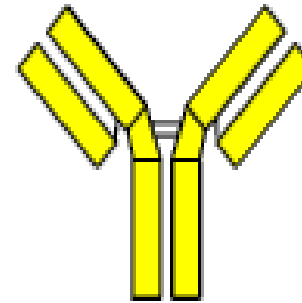
- 1. Inject a mouse with a specific antigen to stimulate its immune system to produce necessary antibodies.**
- 2. Extract mouse spleen cells (containing B-lymphocytes) and culture them in the lab.**
- 3. Extract mouse tumour cells, which grow continuously, and culture them in the lab.**
- 4. Mix spleen cells and tumour cells on the same plate and culture.**

mAb production steps

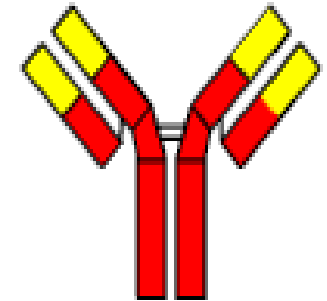
5. Add polyethylene glycol – this causes some B-lymphocytes to fuse with tumour cells to produce a hybrid cell called a hybridoma.
6. Grow the cells under conditions that allow only hybridoma cells to survive.
7. Extract the cells, culture them separately and test the medium around each cell for the specific antibody of interest.
8. Culture the cells making the desired antibody and use as needed.

Types of therapeutically used mAbs:

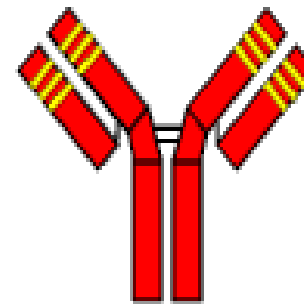
- ✓ Murine mAbs
- ✓ Chimeric mAbs
- ✓ Humanized mAbs
- ✓ fully human mAbs



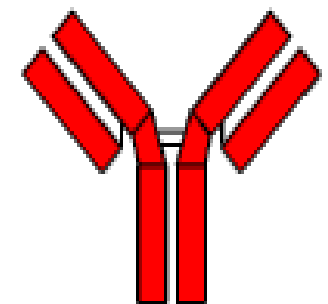
Murine



Chimaeric



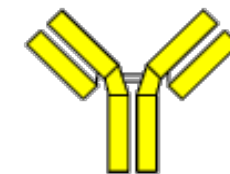
Humanised



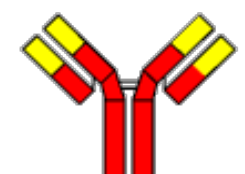
Human

Murine mAbs

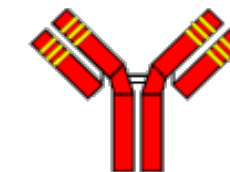
- purified from mouse
 - ✓ Low efficacy
 - ✓ rapid clearance from body due to patients' production of human anti-murine antibodies
 - ✓ drugs of this category have “omab” suffix.
 - ✓ Eg. Tositumomab
 - ✓ Produced by hybridoma technology.



Murine



Chimaeric



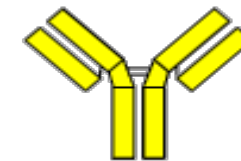
Humanised



Human

Chimeric mAbs

- mAbs containing variable region of mouse antibody and constant regions of human antibody.
- ✓ These antibodies are less immunogenic in man and are much less likely to produce side-effects in human therapy.
- ✓ mAbs of this category have “ximab” suffix.
- Eg. Rituximab.
- Produced by hybridoma and rDNA technology.



Murine



Humanised



Chimaeric



Human

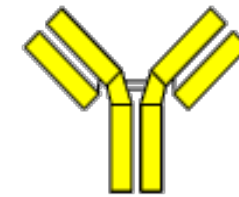
Humanized mAbs

- mAbs containing amino acids of **complementary determining region (hypervariable region) of mouse antibody** and **the rest from human**.

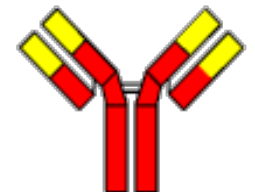
- ✓ antibodies of this category have suffix of “zumab”

Eg. Trastuzumab

- ✓ Produced by hybridoma and rDNA technology.



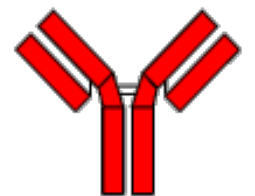
Murine



Chimaeric



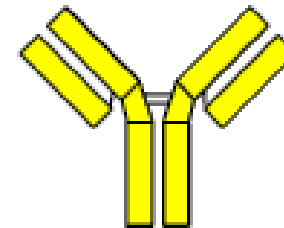
Humanised



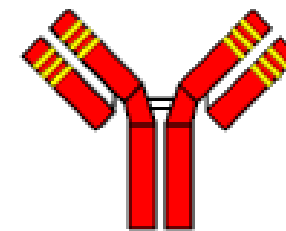
Human

Fully human mAbs

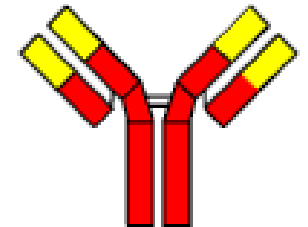
- mAbs produced from human genes.
- ✓ mAbs of this category have suffix of “mumab”
Eg. Adalimumab
- ✓ Produced by **transgenic mouse** which are generated by introducing segment of **Human immunoglobulin loci into the germlines of mice** deficient in mouse antibody production as a result of gene targeting.



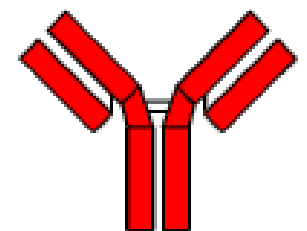
Murine



Humanised



Chimaeric



Human

Applications of mAbs

1. As therapeutic agents
2. As carriers for targeted drug delivery (mAb-drug conjugates)
3. As diagnostic agents

1. Therapeutic applications of mAbs

Mechanism of action of monoclonal antibodies

- Monoclonal antibodies exert their pharmacological effects via multiple mechanisms that include
- **direct modulation of the target antigen,**
- **CDC**
- **ADCC, and**

Therapeutic applications of mAbs.....

MOA. of Monoclonal antibodies....

Direct Modulation of Target Antigen

- Examples of direct modulation of the target antigen include anti-TNF α and anti-IgE that are involved in blocking and/or removal of the target antigen.
- Most monoclonal antibodies act **through multiple mechanisms** and may exhibit cooperatively with concurrent therapies.

Therapeutic applications of mAbs.....

MOA. of Monoclonal antibodies....

Complement-Dependent Cytotoxicity (CDC)

- The complement system is an important part of the innate (i.e., non-adaptive) immune system.
- It consists of many enzymes that form a cascade with each enzyme acting as a catalyst for the next.
- **CDC results from interaction of cell-bound monoclonal antibodies with proteins of the complement system.**
- This cascade spreads rapidly and ends in the formation of the **membrane attack complex (MAC)**, which inserts into the membrane of the target and causes **osmotic disruption and lysis of the target.**

Therapeutic applications of mAbs.....

MOA. of Monoclonal antibodies....

Antibody-Dependent Cellular Cytotoxicity (ADCC)

- The typical ADCC involves activation of NK cells, monocytes or macrophages and is dependent on the recognition of antibody-coated infected cells by Fc receptors on the surface of these cells.
- The Fc receptors recognize the Fc portion of antibodies such as IgG, which bind to the surface of a pathogen infected target cell.
- The Fc receptor that exists on the surface of NK cell is called CD16
- Once bound to the Fc receptor of IgG the **NK cell releases cytokines** such as **IFN- γ** , and **cytotoxic granules** like **perforin** and **granzyme** that enter the target cell and promote cell death.
- This is similar to, but independent of, responses by cytotoxic T-cells.

Therapeutic applications of mAbs.....

MOA. of Monoclonal antibodies....

Apoptosis

- In addition to the above mentioned effector functions they can have direct effects in producing apoptosis or programmed cell death.
- It is characterized by nuclear DNA degradation, nuclear degeneration and condensation, and phagocytosis of the cell remains.

Therapeutic applications of mAbs.....

- Currently, monoclonal antibody therapies are having a significant impact on many disorders including
 - ✓ cardiovascular diseases
 - ✓ rheumatoid arthritis
 - ✓ allergy related asthma
 - ✓ autoimmune disorder
 - ✓ **cancer, etc**

Therapeutic applications of mAbs

- **Cardiovascular diseases**

- **Abciximab**

- ✓ glycoprotein IIb/ IIIa receptor antagonist
- ✓ used in coronary artery procedures(blood thinner)
- ✓ binds to glycoprotein IIb/IIIa receptor on platelets and inhibits interactions of fibrinogen and von Willebrand factor which are crucial in platelet aggregation.

Therapeutic applications of mAbs

- **Rheumatoid arthritis (RA)**
 - **infliximab, adalimumab, golimumab, and certolizumab pegol**
 - ✓ Monoclonal antibodies that bind TNF- α , an inflammatory cytokine
 - ✓ As they bind TNF- α , they inhibit its interaction with TNF receptors and reduce inflammation.
 - ✓ they are also used to treat inflammatory bowel disease

Therapeutic applications of mAbs.....

□ Allergy-related asthma

✓ Omalizumab

- ✓ When IgE binds mast cell, inflammatory mediators such as histamine, cytokines, leukotrienes and chemokines are released and cause broncho-constriction, swelling and mucus production are released into the airways of the lung.
- ✓ Binds directly free IgE antibodies

Therapeutic applications of mAbs.....

❑ Cancer

- mAbs (i.e., immunotherapy) are largely used for treatment of cancer than any other diseases.
- They are used in conjunction with chemotherapy, radiation therapy and surgery.
- Cancer cells carry **specific tumour-associated antigens (TAA)** on their plasma membrane.
- Monoclonal anti-TAA antibodies have been produced.

Therapeutic applications of mAbs.....

- **Breast cancer**
- **Trastuzumab (herceptin)**
 - ✓ Targets human Epidermal Growth Factor receptor-2 (EGFR-2)
 - ✓ EGFR-2 is over expressed in 25-30% of breast cancers
 - ✓ Trastuzumab is thought to act by Interfering in **ligand receptor interaction**.

Therapeutic applications of mAbs.....

■ Colorectal Cancer

○ Panitumumab and cetuximab

- ✓ Both inhibit human Epidermal Growth Factor Receptor 1 (EGFR-1)
- ✓ Arrest cell proliferation,
- ✓ Prevent angiogenesis
- ✓ Promote apoptosis

2. Targeted drug delivery application of mAbs (mAb-drug conjugates)

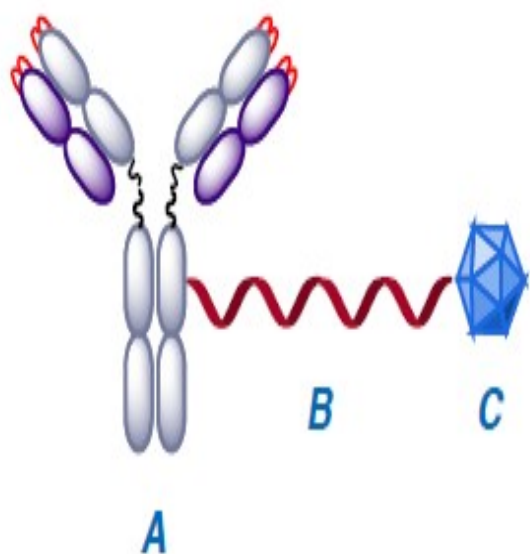
- mAb-drug conjugate utilizes the advantages of mAb **selectively targeting towards antigen-positive cells** and long half-life.
- mAb-drug conjugates are mostly directed to treat cancers.
- Drugs which kill tumour cells or inhibit key proteins in tumour cells are attached to monoclonal anti-TAA antibodies.
- Cancer cells are specifically targeted, avoiding damage to healthy host cells.

Targeted drug delivery (mAb-drug conjugates.....)

Components of mAb-drug conjugates

An integrated system of mADC consists:

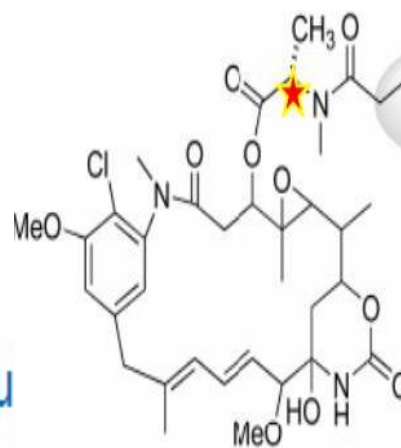
- Targeting antibody (e.g., specific to tumor antigen)
- Drug molecule
- Linker



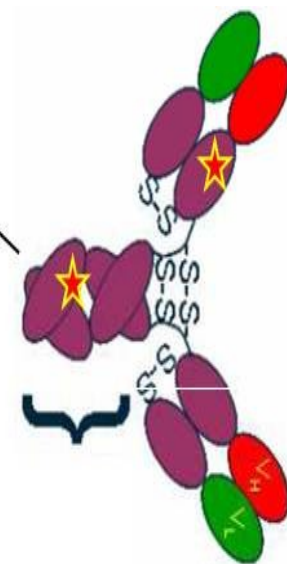
A. Antibody

B. Linker

C. Small-molecule

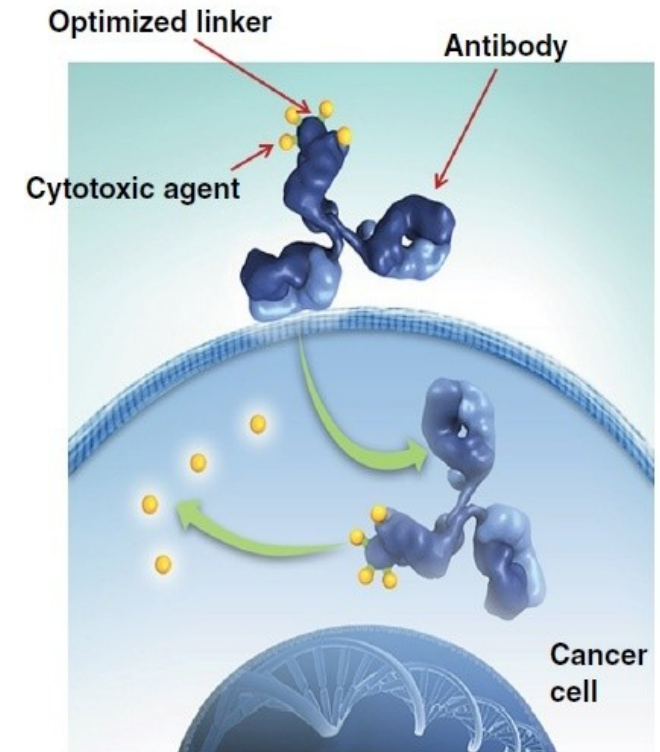


SPDB
(disulfide
linker)



2.Targeted drug delivery / General mechanism of action of mAb-drug conjugates in cancer

- Binds to target on surface of cancer cell
- Internalized into the cell
- Cytotoxic agent is released inside the cell
- Cytotoxic agent affects its target (eg., DNA, tubulin etc)
- Cell death
- Eg. trastuzumab emtansine (T-DM1)



3.Diagnostic application of mAbs

□ Immunoassays using ELISA

- ELISA (enzyme linked immuno sorbent assay) exploits the use of an enzyme attached to one of the reagents utilized in the test.
- Antibodies are fundamental reagents in ELISA.

Basic principles of ELISA test



Where, I= immobilizing solid phase

Ag = antigen Ab*E= Antibody labeled with enzyme

W= wash += addition S= substrate

Diagnostic application of mAbs.....

Basic principles in ELISA test

- **Steps in the assay process:**

Step i: passive adsorption of antigen to plate surface by incubation in defined buffer

Step ii: wash

Step iii: addition and incubation of enzyme-labeled antibody

Step iv: wash

Step v: substrate addition and incubation of color development system

Step vi: read

Diagnostic application of mAbs.....

- Subsequent addition of relevant enzyme substrate/ chromogen causes a color change.
- The result can be read by eye or quantified using specially designed spectrophotometers.